



**WAN**

Connects remote LANs via S/Ps for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**

- Bandwidth:** Plan for normal load, peak usage, reserved capacity for VoIP
- Control/Security:** Reliance on the **provider** → encryption and segmentation
- Availability:** Use redundancy, diverse paths, SLAs to reduce outage impact
- Management:**
  - In-band:** Manage devices over the production WAN
  - Out-of-band:** Use separate management path for recovery/emergencies

**PRIVATE WAN**

**Point-to-Point:** Leased L2 line (Ethernet), monthly fee; private circuit, fast  
**Dark Fiber:** Physical fiber lease; costly; ISPs prefer selling lambdas  
**Coarse Wavelength Division Multiplexing (CWDM):** x16, cheaper, 120km, passive MUX, use mirrors to aggregate/separate signals (analog)  
**Dense Wavelength Division Multiplexing (DWDM):** x80, 10GB/s, 1000km, active MUX, more precise, uses software ("smarter"), wideband  
**Connection-oriented:** Predelineated path, packet IDs (ATM, Frame Relay)  
**Connectionless:** No setup, full address in each packet (Ethernet, MPLS VPN)  
**Circuit-Switched:** Requires signaling protocol to establish e2e circuit  
**Packet-Switched:** Expects network to know how to forward packets

**PUBLIC WAN**

**Site-to-Site VPN:** Fixed locations, devices locked to IP addr  
**Remote Access VPN:** Changing locations, devices not locked to IP addr  
**Software-Defined WAN (SDWAN):** connects LANs across large distances using controlling software that works with a variety of networking hardware.

**MULTI PROTOCOL LABEL SWITCHING (MPLS)**

- Advantage of eBGP at PE-CE: No mutual redistrib., same routing process
- Control plane (e.g. OSPF) → to learn labels
- iBGP to exchange NLRI (RD/RT/IPv4 Pref, NextHop&VPN Lb) between PE
- Traffic encrypted flowing over MPLS L3VPN backbone

**Label Switched Router (LSR):** Forwarded labeled packets  
**Label Switched Path (LSP):** Pre-determined path across MPLS network  
**Unicast Reverse Path Forwarding (uRPF):** checks source of each packet & verifies that source is in routing table

**imp-nul:** networks are directly connected, no more label switching  
**Virtual Private Wire Service (VPWS):** Pseudowire (PW); No MAC learning at PE level; P2P L2 VPN, allows two sites to connect as if directly linked  
**Virtual Private LAN Service (VPLS):** Emulates LAN segment across MPLS backbone, provides multipoint L2 connectivity; MAC learning at the PE level

**CONFIG**

1) RIB entry to 10.3.3.0/24 lists a next-hop IP address of 192.168.12.2

2) PE1 compares that entry to the list of interface IPs on each peer and finds nei with IP 192.168.12.2

3) PE1 notes that for 10.3.3.0, LIB contains one local and two remote labels

4) Among the known labels, one was learned from 2.2.2.2, with label 22

PE1# show ip route 1 inc 10.3.3.0

1 10.3.3.0 [90/2835456] via 192.168.12.2, 00:05:08, Serial0/0/1

PE1# show mpls ldp bindings 10.3.3.0 24

lib entry: 10.3.3.0/24, rev 28

Local bindings: Lsr: 2.2.2.2, 0, label: 22

remote binding: Lsr: 192.168.12.2, 4.4.4.4, 0, label: 86

PE1# show mpls ldp nei 1

Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0

TCP connection: 2.2.2.2.2.34326 -> 192.168.12.2.1.1.1.646

State: Oper; Hgss sssent/rcv: 37/33; Downstream

2 Up time: 0:02:57

LDP discovery sources:

Serial0/0/1, Src IP addr: 192.168.12.2

Addresses bound to LDP Ident:

192.168.12.2, 2.2.2.2, 192.168.12.0, 192.168.12.4

192.168.12.2

**ROUTER TYPES**

**Customer Edge (CE):** No knowledge of MPLS, no labels; connected to **Provider Edge (PE):** On edge of MPLS net; runs BGP & LDP; uses VRFs  
**Provider or LSR (P):** Inside MPLS VPN, no CE connection; forwards labels

**DATABASES, PLANES**

**Control Plane:** Builds routing/labeling (LIB, LIB)  
**Data Plane:** Forwards packets (FIB, LFIB); pushes/swaps/pops labels  
**Class of Service Forwarding (CoS):** (Cisco) LFIB without labels → FIB?  
**Routing Information Base (RIB):** Learned prefixes from routing protocols  
**Forwarding IB (FIB):** Built from RIB; only best routes for forwarding  
**Label IB (LIB):** All label mappings; No BGP prefixes; 1 label per prefix  
**Label Forwarding IB (LFIB):** Built from LIB; used for actual forwarding decisions. (LFIB only contains currently best LSP decision: Routing Protocol)

**MPLS HEADER**

4-byte header before IP. With MPLS VPN, there will be a stack of two headers  
**Label IB:** active MPLS labels  
**EXP (3b):** QoS/CoS, Now called Traffic Class (TC)  
**S-bit (1b):** Bottom of label stack indicator, 1=True = last label before IP header  
**TTL (8b):** time-to-live (equal to IP TTL)

**TTL**

- Ingress PE decrements IP TTL field & copies IP TTL field into new MPLS TTL
- P routers decrements MPLS TTL
- Egress PE decrements MPLS TTL, pops final MPLS header, copies IP TTL
- If provider doesn't want to expose MPLS network: disable **MPLS TTL propagation** on PE, PE sets MPLS TTL=255, egress PE leaves original IP TTL unchanged → new network appears as single router hop from TTL perspective

**LABEL DISTRIBUTION PROTOCOL (LDP) - CONTROL PLANE**

- Distributes labels to neighbors, triggered by new IP route in FIB
- Hello messages** sent via UDP (Port 646) to 224.0.0.2 to discover neighbors
- TCP (646) connection used to exchange label bindings (prefer to control lab)
- Router advertises all local bindings after TCP session is up
- Label mapping used to build LIB → LFIB
- LDP router ID must be reachable (via routing table)
- Each router manages local labels independently
- Relies on underlying routing protocol for best route & loop prevention

**MPLS L3 DATA PLANE**

**Outer label:** Transport label (LDP); identifies LSP between ingress/egress PE  
**Inner label:** Transport label (4b-BGP); identifies customer VRF  
**Push:** Ingress PE; classify and label packets  
**Swap:** P router; replaces label, forwards based on new VRF  
**Pop:** Egress PE removes label; sends original packet to CE  
**Penultimate Hop Popping (PHP):** penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

**VIRTUAL ROUTING AND FORWARDING (VRF) TABLES**

- Virtual router inside a PE. Maintains isolated RIB + FIB and separate routing process per CE (VPN isolation)
- Exists per MPLS-aware PE router; one per attached customer

**ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)**

**Route Distinguisher (RD):** *Uniquely identifies* VPN routes (per VRF) → allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN.  
**Example:** VPN4w NLRI (MP-BGP): RD-IPv4 prefix (12 bytes: RD 8b, IP 4b)  
**Route Target (RT):** Controls route **import/export** between VRFs. Used as **extended BGP community path attributes**. Typically in the form **ASN:nn** or **IP:nn**, e.g., 65000:100, 1:10

**How RTs Work:** Route is tagged with an RT when advertised by BGP. Other VRFs import the route if the RT matches their import policy. Allows overlapping or shared connectivity between tenants (e.g., shared services).

**OVERLAY TECHNOLOGIES**

**Overlay network** (virtual) sits on top of the **underlay network** (physical).  
**Generic Routing Encapsulation (GRE):** Encapsulates data packets, unencrypted, P2P. Enables usage of unsupported protocols. Carries passenger protocol/original packet inside carrier/transport IP header routed by underlay

**SEGMENT ROUTING (SR)**

**Source routing:** Sender defines full path using Segment (= Instruction) List  
**Segment Identifier (SID):** Each SID = 1 instruction (e.g., forward via ECMP, specific face, or to a service)  
**Segment List:** Ordered SID list carried in packet header; defines full route  
**State in packet:** No per-flow state, intermediate nodes follow SID instructions  
**No new control plane:** Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed  
**Data plane:** Can operate over MPLS or IPv6  
**Simple but powerful:** Enables TE, fast reroute, policy routing

**SEGMENT LIST OPERATIONS**

**Push:** Insert SIDs into packet; set active SID (top of list)  
**Continue:** Active SID not yet completed; keep processing it  
**Next:** Current SID completed; activate next SID in list

**SEGMENT SIGNIFICANCE**

**Global Segments:** Known and supported by all SR nodes in the domain, installed in forwarding tables across the network (e.g., "Forward packet according to shortest path to Node1"). Defined in the SR Global Block (SRGB) and should be consistent across all nodes  
**Local Segments:** Defined and installed only on originating node, not forwarded by others, but must be understood network-wide (e.g., "Forward packet on interface to Node2"). Defined in the SR Local Block (SRLB) and are specific to the local SR node.

**CONTROL PLANE SEGMENT TYPES**

**IGP Prefix Segment:** Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries: "steer traffic along ECMP-aware shortest path"  
**IGP Node Segment:** Global SID for a specific node (shortest-path forwarding)  
**IGP Next Hop Segment:** Local SID for a specific next hop; traffic sent to nearest IGP Adjacency Segment  
**Local SID:** direct link to neighbor  
**L2 Adjacency SID:** Local SID for Layer-2 segment (e.g., Ethernet link)  
**Combining Segments:** End-to-end paths can mix IGP and BGP segments. Traffic to BGP Anycast → more ECMP in data centers

**SR-MPLS**

- Reuses existing MPLS data plane → no hardware change needed
- Segments = MPLS labels; Segment List = label stack (top = active)
- Segments distributed via IGP/BGP; no LDP necessary (but possible)
- Supports both IPv4 and IPv6 networks
- PUSH= PUSH, CONTINUE= SWAP, NEXT= POP

**BENEFITS OF SEGMENT ROUTING**

**Benefits:** Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust  
**Source Routing:** Labels distributed inline with centralized optimization  
**Traffic Engineering (TE):** Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS

**DRAWBACKS OF TRADITIONAL NETWORKS**

**Control Plane:** LDP/RSVP-TE adds complexity  
**Scalability:** Per-flow/path limits growth; LSP, signaling overhead increase fast

**VIRTUAL EXTENSIBLE LOCAL AREA NETWORK (VXLAN)**

Data plane technology which encapsulates Ethernet frames in UDP data packets to tunnel layer 2 frames over a layer 3 network. The underlay network is unaware of the overlay.  
**Issues of L2: STP:** Max amount of VLANs (4094), Large MAC Address tables  
**VXLAN:** Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation.  
**VXLAN Network Identifier (VNI):** 24-bit identifier (Up to 16 million segments) that defines the VXLAN broadcast domain.  
**Virtual Tunnel Endpoint (VTEP):** Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.  
**Network Virtual Interface (NVE):** Logical interface on a VTEP used for VXLAN tunnel operations.

**FRAME FORMAT**

- Original Ethernet frame → VXLAN Header → UDP → Outer IP Header
- The VXLAN header contains the 24-bit VNIID and flags.
- Outer headers allow Layer 2 frames to traverse IP underlay networks.

**VIRTUAL NETWORK IDENTIFIER (VNI)**

- 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
- Replaces traditional VLAN IDs (12-bit)
- Used by VTEPs to map traffic into corresponding Layer 2 domains.

**VXLAN TUNNEL ENDPOINT (VTEP)**

- Connects the overlay (VXLAN) and underlay (IP) networks. A VTEP can have multiple VNI interfaces, but they associate with the same VTEP IP interface.
- Software VTEP:** Located on hypervisors using virtual switches.
- Hardware VTEP:** Located on routers/switches with ASICs for performance.
- VTEP Interface:** Connects to the underlay network, handles encapsulation.
- VNI Interface:** Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

**MAC ADDRESS LEARNING**

- Each VTEP maintains a VXLAN mapping table linking destination MAC addresses to remote VTEP IPs.
- Data plane:** reactive, Flood and learn, MAC learned from incoming traffic, BUM via ingress replication or multicast underlay. No control-plane needed
- Control plane:** proactive, MP-BGP/IPv6/MAC-to-VTEP advertised via BGP. No flooding needed for known MACs, ARP suppression, silent host handling
- Flood and learn:** Host H1 sends ARP request, switches learn H1's MAC. ARP request is flooded to H2 → H2 responds; switches learn H2's MAC.
- Static VXLAN:** Manual MAC-to-VTEP mappings. Sends copy of BUM traffic to all participating VTEPs. Doesn't scale well; BUM traffic is inefficient.
- Multicast VXLAN:** VTEPs join multicast groups per VNI. Scales better, offloads BGP replication. 20+ VTEPs = too much traffic, doesn't scale well
- MP-BGP EVPN:** Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

**ETHERNET VIRTUAL PRIVATE NETWORK (EVPN)**

Overcome flood-and-learn limitations, doesn't rely on data plane learning, utilizes router control plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, L2 and L3 Support.

**LAYER 2**

L2 bridging across L3 networks  
**BGP Control Plane:** Distributes MAC info (no flooding → efficiency)  
**VXLAN Overlay:** Encapsulates L2 in L3 UDP (data plane)  
**Multi-Tenancy:** via VNI segmentation  
**Redundancy:** All-active multihoming, ECMP, fast convergence

**USE CASES**

- Multi-tenant datacenter interconnections (DCI)
- Extending L2 over WAN between remote sites
- Scalable, segmented L2 fabrics

**AUTODISCOVERY VIA ROUTE REFLECTORS**

- RR reflects EVPN routes to other PEs
- RR doesn't participate in EVPN or pseudowires
- RR needs only **address-family l2vpn evpn**
- L2VPN RIB stores endpoint/VFI info for control plane
- BGP UPDATE from spines contains **ORIGINATOR\_ID** (origin leaf)

**HOST DETECTION**

- Host connects to VTEP → MAC learned locally
- VTEP advertises MAC + L2VNI via BGP EVPN
- MAC learning follows normal Ethernet semantics

**INGRESS REPLICATION (IR)**

- Forwarding BUM traffic. When Multicast underlay network is not used, IR handles multi-destination traffic (ARP → unicast). Ingress device replicates BUM packets, sends them as separate unicast to egress devices.

**EARLY ARP TERMINATION (ARP SUPPRESSION)**

- Avoids flooding ARP requests
- VTEP queries control plane for MAC/IP/VNI mapping
- If known → direct unicast (no broadcast)

**Silent Host Flow (Fallback)**

- If IP/MAC unknown → ARP sent via **ingress replication**
- Replicated ARP request goes to remote VTEPs
- Only correct host responds → update reflected to all VTEPs
- Future traffic uses updated BGP mapping

**LAYER 3**

- By adding L3 features, data routing efficiency + smooth connections improve

**EVPN ROUTE TYPES**

- (1) Ethernet Auto-Discovery Route (3) Inclusive Multicast Route (4) Ethernet Static Route (5) Selective Multicast Ethernet Tag Route (7) IGMP Join Synchrony Route (8) IGMP Leave Synchrony Route

**Type 2 – Host Advertisement:** Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when host connects to VTEP.

**[2]: Route Type [0]: Eth. Segment ID [0]: Eth. Tag ID [48]: MAC Addr len [a2b9.4605.948d]: MAC [32]: IP Addr len [10.1.1.1]: IP**

**RT: L2VNI (VLAN) RT: L3VNI (VRF) Overlay Encapsulation: 8 - VXLAN L2VNI L3VNI Router MAC of Remote VTEP Remote VTEP IP**

**LEAF-S5 show bwp l2vpn evpn 10.1.1.1**

**BGP routing table information for VRF default, address family L2VPN EVPN**

**Route Distinguisher: 10.0.0.3:32777**

**BGP routing table entry for [2]:[0]:[0]:[48]:[a2b9.4605.948d]:[32]:[10.1.1.1]/772, version 424**

**Paths: (2 available, best #2)**

**Flags: (0x00020) on xmit-1st, is not in L2rib/evpn, is not in HW**

**Advertised path-id 1**

**Path type: Internal, path is valid, is best path, no labeled nexthop**

**AS-Path: NONE, path sourced internal to AS**

**10.0.1.101 (metric 81) from 10.0.0.1 (10.0.0.1)**

**Origin IGP, MED 0, LocalPref 100, weight 0**

**Received label 100010 100999**

**Extcommunity: RT:645000:100010 RT:645000:100999 ENCAP:8**

**Router MAC:0000.0000.0100**

**Originator: 10.0.0.3 Cluster List: 10.0.0.1**

**Type 5 – Subnet Advertisement:** Advertises IP prefix + prefix length to L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.

**[5]: Route Type [0]: Eth. Segment ID [0]: Eth. Tag ID [24]: IP Pref. len [198.51.100.0]: IP Prefix [6.0.0.0]: GW IP**

**L3VNI RT: L3VNI (VLAN) Overlay Encapsulation: 8 - VXLAN Router MAC of Remote VTEP Remote VTEP IP**

**LEAF-S5 show bwp l2vpn evpn route-type 5**

**BGP routing table information for VRF default, address family L2VPN EVPN**

**Route Distinguisher: 10.0.0.3:3**

**BGP routing table entry for [5]:[0]:[0]:[24]:[198.51.100.0]:[6.0.0.0]/224, version 6421**

**Paths: (2 available, best #2)**

**Flags: (0x00002) on xmit-1st, is not in L2rib/evpn, is not in HW**

**Advertised path-id 1**

**Path type: Internal, path is valid, not best reason: Neighbor**

**Address: no labeled nexthop**

**AS-Path: NONE, path sourced internal to AS**

**203.0.113.1 (metric 81) from 192.0.2.12 (192.0.2.2)**

**Origin incomplete, MED 0, LocalPref 100, weight 0**

**Received label 10001**

**Extcommunity: RT:65000:10001 Router MAC:0000.c608.1008**

**Originator: 192.0.2.3 Cluster List: 192.0.2.2**

**HOST DETECTION**

- Host sends ARP/ND to local VTEP
- VTEP learns MAC/L2VNI and IP/L3VNI
- Info is advertised in EVPN (control plane)

**HOST DELETION & EVPN**

**Host Deletion:** When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.  
**Host Move:** When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

**VIRTUAL ROUTING AND FORWARDING (VRF)**

- Supports independent policies per tenant
- Key for scaling and multi-customer separation

**DISTRIBUTED ANYCAST GATEWAY (DAG)**

- Same gateway IP+MAC on all VTEPs

**Enables local default gateway for hosts in MPLS network with IRB**

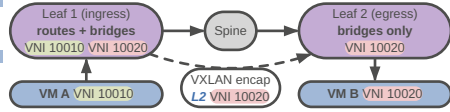
**Supports mobility + optimal forwarding across BGP EVPN fabric**

**INTEGRATED ROUTING AND BRIDGING (IRB)**

- Enables inter-VLAN routing inside EVPN
- Avoids central gateway → "no traffic tromboning"

**ASYMMETRIC IRB (L2)**

- Routing only ingress, bridging on egress
- VXLAN uses destination VNI in both directions
- One L2 VNI per VLAN/Subnet
- Simple config, but requires all VLANs/VNIs on all VTEPs (scaling issues)



1) VM A → Leaf 1 in VNI 10010 (frame to anycast gateway)

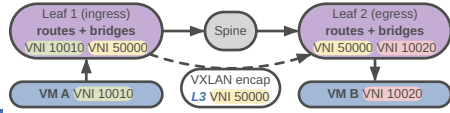
2) Leaf 1 routes locally. VNI 10010 → VNI 10020 (L3 lookup, MAC rewrite)

3) Leaf 1 encapsulates with destination L2 VNI 10020 → spine → Leaf 2

4) Leaf 2 decapsulates and bridges in VNI 10020 to VM B → no routing

**SYMMETRIC IRB (L2 + L3)**

- Routing/bridging on ingress + egress VTEPs
- Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
- Scales well; clean separation of MAC and IP
- Leaf 1 and Leaf 3 connect to the same destination VTEPs for Type 2 routing (Done by configuring Type 5 routing)



1) VM A → Leaf 1 in VNI 10010 (frame to anycast gateway)

2) Leaf 1 routes locally into the L3 VNI (tenant VRF transit)

3) Encapsulation carries the L3 VNI 50000 → Leaf 2

4) Leaf 2 decapsulates, looks up dest IP in the VRF, routes into VNI 10020

5) Leaf 2 bridges to the destination L2 VNI on the subnet

**MULTI-PROTOCOL BGP (MP-BGP)**

**Address Family Identifier (AFI):** Category of information being carried  
**Subsequent AFI (SAFI):** Narrows down the specific type within that category  
**EVPN NLRI:** Uses MP-BGP with specific AFI/SAFI. Supports multiple route types and attributes, unsupported routes are dropped by BGP

- Enables protocol-based VTEP discovery and host reachability via control-plane learning. Reduces flooding by replacing data-plane learning
- Extends BGP with multiprotocol capabilities (AFI/SAFI)
- Uses **MP REACH NLRI** and **MP UNREACH NLRI** for route advertisement and withdrawal (advertises VPNv4 between PEs in MPLS)
- Uses the **VPNv4 NLRI** to distinguish between duplicate IPv4 prefixes

**BGP CONTROL PLANE**

- PEs learn MACs from local CE's (data plane)
- MACs advertised via BGP (control plane)
- Uses Route Distinguishers and MPLS labels
- Remote PEs update L2 RIB/FIB with MAC and next-hop info
- Enables seamless L2 across IP/MPLS backbone

**CDN**

**Origin Server:** Central content source (original files), usually in a datacenter  
**Edge / CDN Server:** Also called **Point of Presence (POP)**. Geographically distributed, cache content

**Embedded POP:** Deployed directly inside ISP's local network  
**DNS Infrastructure:** Directs users to optimal edge server (e.g. Geo-Routing)

**KEY BENEFITS**

**Latency Reduction:** Nearby edge servers reduce round-trip time  
**Availability:** Failover and redundancy in case of node failure  
**Scalability:** Handles traffic spikes via load balancing  
**Cost Optimization:** Reduces backend and transit load on origin  
**DDoS Protection:** Edge servers absorb attacks → not all traffic on one server  
**Global Load Reduction:** Less long-distance traffic across the Internet

**REQUEST ROUTING TECHNIQUES**

- Decides which edge server should serve a client request
- Goal: Best performance (e.g. proximity, load, responsiveness)

**DNS-BASED GEO-ROUTING**

- Each edge has a unique IP
- DNS server picks closest/optimal edge server based on:
  - Resolver IP location (not user) → can cause wrong choice
  - DNS (MaxMind, IP2Location), load, latency, business rules

**EDNS0 (I) and Client Subnet Extension (ECS):**

- Resolver includes part of client IP in DNS request (e.g. /24 subnet)
- Authoritative DNS makes better decision based on actual client region
- Improves accuracy without revealing full IP, increases caching complexity

**ANYCAST WITH BGP**

- Same IP (e.g. 7.7.7.7) advertised from multiple locations
- BGP routing decides which path is "best" (AS-path, local pref, etc.)
- No DNS logic or per-client decision – pure BGP convergence
- Pros:** Fast failover, simple, no app logic needed
- Cons:** Control BGP, BGP + best latency, route flapping risk

**MANAGING TRAFFIC**

**Anycast:** Modify BGP attributes/withdraw IP prefix. Not precise  
**Software:** Shifting via Encapsulation tunnels (GRE)

**LOAD BALANCING**

**Load balancing device:** Efficient, reliable, High cost, single point of failure  
**ECMP:** ToR router balances traffic. Simple, cheap, Adding/removing server changes hashing pool, re-hashing breaks TCP connections if: GRE tunnel

**HTTP CACHING & HEADERS**

**Caching:** is controlled via HTTP headers between clients, proxies, and servers  
**Cache-Control:** Main directive (**no-cache, no-store, max-age, must-revalidate, etc.**)

**Expires:** Absolute expiration time (older method, replaced by **Cache-Control**)  
**ETag:** Validator tag (version/hash), used with **If-None-Match**  
**Last-Modified:** Timestamp used with **If-Modified-Since** for validation  
**Age:** Time (in seconds) since response was fetched from origin  
**Validation:** Client uses **ETag** or **Last-Modified**; server → 304 if unchanged

**QoS**

**Internet is best effort:** No guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization  
**Quality of Experience (QoE):** Perceived service quality from user perspective  
**Route Pinning:** Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)  
**Routing-based:** Path selection; QoS-aware routing; Choose best path  
**Node-based:** Packet treatment; Congestion mgmt; queuing/scheduling/policing/shaping; "What happens to packets at each hop?"  
**PAK-Priority:** System queues (routers) that prioritizes routing protocol packets

**4-Class Model**

| 4-Class Model        | 8-Class Model     | 12-Class Model          |
|----------------------|-------------------|-------------------------|
| Real Time            | Voice             | Voice                   |
|                      | Interactive Video | Real-Time Interactive   |
|                      | Streaming Video   | Multimedia Conferencing |
|                      |                   | Broadcast Video         |
|                      |                   | Multimedia Streaming    |
| Signaling or Control | Signaling         | Signaling               |
|                      | Network Control   | Network Control         |
|                      |                   | Network Management      |
| Critical Data        | Critical Data     | Transactional Data      |
|                      |                   | Bulk Data               |
| Best Effort          | Best Effort       | Best Effort             |
|                      | Scavenger         | Scavenger               |

**NETWORK PERFORMANCE METRICS**

**Bandwidth [Gbit/s]:** Maximum transfer capacity of a link (width of pipe)  
**Throughput [Gbit/s]:** Successfully delivered data (water that reached dest)  
**Latency / Delay [ms]:** Time for packets to travel src → dest (length of pipe)  
**End-to-End Delay:** Total time sender to receiver  
**One-Way Delay:** From first bit sent to last bit received  
**Components:** Transmission delay (time to push onto link), **Processing delay** (lookup, queuing), **Propagation delay** (physical travel time)  
**Jitter [ms]:** Variation in delay between packets, caused by re-routing/queuing (Voip>30ms), Calc: highest - lowest delay. Doesn't impact TCP  
**Packet Loss Ratio (PLR) [%]:** Dropped packets due to congestion or errors

**QUEUEING**

**Incoming buffer:** Usually only transit, cannot be filled up in case of a CPU overload; leads to drops  
**Outgoing buffer:** Filled up with packets waiting for transmission, queue management is focused on the outgoing buffer

**ALGORITHMS**

**First-In-First-Out (FIFO):** Basic, no prioritization  
**Priority Queuing (PQ):** Multiple queues, serve highest first; others may starve  
**Round-Robin (RR):** One packet per queue in turn (fair, but ignores priority)  
**Weighted Fair Queuing (WFQ):** RR with weights, n packets per queue  
**Class-Based WFQ (CBWFQ):** WFQ with user-defined classes (logical queues), queues limits, max bandwidth guaranteed or max % of bandwidth  
**Low Latency Queuing (LLQ):** Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol...)

**QUEUE MANAGEMENT**

**tail Drop:** Drops packets when queue full; huge interruption of traffic → same as no connectivity  
**TCP Global Sync:** Many TCP flows back off and restart simultaneously (AIMD) → link underutilization  
**TCP Starvation:** TCP slows down after drops (congestion control), UDP doesn't → queues filled with UDP TCP squeezed out  
**Random Early Detection (RED):** Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their window sizes  
**Weighted RED (WRED):** RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic, max thresh → all packets dropped

**POLICING VS. SHAPING**

**Policing (Inbound mostly):** Drops packets that exceed configured rate limits  
**Shaping (Outbound):** Buffers packets to smooth traffic bursts

**QoS MODELS**

**BEST EFFORT**  
No guarantees, all traffic treated equally (follow Internet neutrality)  
**INTEGRATED SERVICES (INTSERV)**  
→ End-to-end QoS (guaranteed for specific apps)  
→ Per-flow resource reservation  
→ Precise but not scalable (uses RSVP)  
→ App informs network of traffic traits, requests service before sending data

**DIFFERENTIATED SERVICES (DIFFSERV)**  
→ Scalable approach using marking. No hard guarantees, Cost-effective  
→ Divides traffic into classes based on business requirements  
→ Routers are configured with policy on how to prioritize classes

**Marking (DSCP)**  
**Differentiated Services Code Point (DSCP):** 6-bit field in the IP header used to classify and manage network traffic  
**L3 Marking:** ToS byte → DSCP (6 bits) + IP Precedence (3 bits) [overlapping]  
**DSCP:** (6-bit in IP header) marks packets for QoS; classifies traffic  
**L2 Marking:** Dot1q header → 802.1p/CoS bits  
**EXP:** (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP

**Behavior (PHB)**

**Per-Hop Behavior (PHB):** Defines what routers actually do  
**Class Selector (CS):** Compat with old IP Precedence, simple prio levels  
**Assured Forwarding (AF):** Multiple classes + drop priorities  
**Expedited Forwarding (EF):** Very high prio, e.g. VoIP

**MODULAR QoS CLI (MQC)**

**Class Map:** Define traffic classes (e.g., match voice or video)  
**Policy Map:** Define actions for each class (e.g., limit, shape, priority)  
**Service Policy:** Apply policies to interfaces or directions (in/out)

**OTHER INFOS**

**AD:** Inter-protocol choice (e.g., OSPF vs RIP) → lower wins  
**Cost/Metric:** Intra-protocol choice (e.g., OSPF path A vs B) → lower wins  
**Routing Preference Order (across protocols):**

- Most specific prefix
- Lowest Administrative Distance
- Static default route

**Administrative Distances:** (Smallest Administrative Distance wins)

|                   | Protocol | Distance |
|-------------------|----------|----------|
| Connected         | 0        |          |
| Static (next hop) | 1        |          |
| BGP External      | 20       |          |
| OSPF Internal     | 90       |          |
| EIGRP             | 110      |          |
| ISIS              | 115      |          |
| RIP v1/v2         | 120      |          |
| EIGRP External    | 170      |          |
| BGP Internal      | 200      |          |

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